

Input and Output Screenshots for Week 5 API Assignment

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Input:

```
1  #=====
2  #Title: Week 5 Exercise
3  #Author: Wade Eckert
4  #Date: 06 July 2025
5  #Modified by: Wade Eckert
6  #Description: This Program demonstrates an example of how to
7  #use the requests module to get data from OpenWeather using an API key.
8  #It also demonstrates how to use the json module to parse the data
9  #returned from the web service.
10 #This program uses the API key 906b6939735602a519447e37a839d229 to access
11 #the OpenWeather map web service.
12 #=====
13
14 #Import the necessary modules
15 import requests
16 import json
17 import re
18
19 #Define the function to get the weather data from the API and parse the JSON response
20 def get_weather(zip_code):
21     url = f"http://api.openweathermap.org/data/2.5/weather?zip={zip_code},us&appid=906b6939735602a519447e37a839d229"
22     response = requests.get(url)
23     return json.loads(response.text)
24
25 #Define the main function to run the program
26 def main():
27     while True:
28         zip_code = input("Enter your zip code: ")
29         print()
30
31         #Validate the zip code using RegEx
32         if not re.match(r'^\d{5}$', zip_code):
33             print("Invalid zip code.")
34             print()
35             continue
```

```

36
37 #Get the weather data from the API and parse the JSON response to display the weather forecast
38 data = get_weather(zip_code)
39 #Check if the zip code is valid
40 if data['cod'] == '404':
41     print("Zip code not found.")
42     print()
43     continue
44 #If the zip code is valid, display the weather forecast
45 else:
46     print(f"Weather forecast for {data['name']}, {data['sys']['country']}:")
47     #Convert the temperature from Kelvin to Fahrenheit
48     fahrenheit = (data['main']['temp'] - 273.15) * 9/5 + 32
49     print(f"Temperature: {fahrenheit:.2f} F")
50     print(f"Weather: {data['weather'][0]['description']}")
51     print(f"Humidity: {data['main']['humidity']}%")
52     print(f"Wind Speed: {data['wind']['speed']} m/s")
53     print()
54
55 #Make a loop to ask the user if they want to enter another zip code
56 while True:
57     again = input("Do you want to enter another zip code? (y/n): ")
58     print()
59     if again.lower() == 'n':
60         return
61     elif again.lower() == 'y':
62         break
63     else:
64         print("Invalid input.")
65         continue
66

```

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66
67 #Define the intro function to display the intro message
68 def intro():
69     print("Welcome to the Weather Forecast Program!")
70     print("This program will display the weather forecast for the zip code you enter.")
71     print("Please enter a valid zip code to continue.")
72     print()
73
74 #Define the outro function to display the outro message
75 def outro():
76     print("Thank you for using the Weather Forecast Program!")
77     print("Goodbye!")
78
79 #Call the intro function to display the intro message
80 intro()
81
82 #Call the main function to run the program
83 if __name__ == '__main__':
84     main()
85
86 #Call the outro function to display the outro message
87 outro()

```

Output:

```
Welcome to the Weather Forecast Program!  
This program will display the weather forecast for the zip code you enter.  
Please enter a valid zip code to continue.
```

```
Enter your zip code: 80002
```

```
Weather forecast for Arvada, US:  
Temperature: 63.95 F  
Weather: clear sky  
Humidity: 60%  
Wind Speed: 3.6 m/s
```

```
Do you want to enter another zip code? (y/n): y
```

```
Enter your zip code: 47546
```

```
Weather forecast for Jasper, US:  
Temperature: 73.63 F  
Weather: clear sky  
Humidity: 100%  
Wind Speed: 0 m/s
```

```
Do you want to enter another zip code? (y/n): n
```

```
Thank you for using the Weather Forecast Program!  
Goodbye!
```

```
□
```